MALARIA TRENDS IN ZAMBIA (2015 TO 2025)

A Data Analysis Report

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Data Source: Simulated based on WHO and NMEC trends

EXECUTIVE SUMMARY

This report analyzes malaria trends in Zambia from 2015 to 2025, examining provincial distribution, age group vulnerability, and population correlations. Key findings show a 32% reduction in national malaria cases, with children under five remaining the most vulnerable group. Urban provinces like Lusaka and Copperbelt have significantly lower malaria rates due to better healthcare access and intervention coverage.

1. INTRODUCTION

Malaria remains one of the most significant public health challenges in sub-Saharan Africa, contributing to high rates of morbidity and mortality, particularly among vulnerable populations such as children under five and pregnant women.

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According to the World Health Organization, the African region accounts for approximately 95% of all malaria cases and 96% of all malaria deaths globally. Despite substantial investments in malaria control and elimination efforts over the past two decades, the burden of the disease remains unevenly distributed, with some countries and regions making remarkable progress while others continue to struggle.

Zambia, like many of its neighbors, has been implementing a range of malaria control interventions, including insecticide-treated bed nets, indoor residual spraying, seasonal malaria chemoprevention, and improved case management through rapid diagnostic tests and artemisinin-based combination therapy. The National Malaria Elimination Centre (NMEC) has been coordinating these efforts with support from international partners such as the World Health Organization, the Global Fund, and the United States President's Malaria Initiative.

However, despite these efforts, the burden of malaria varies significantly across the country. Some provinces have achieved near elimination, while others continue to report high incidence rates year after year. Understanding these disparities is essential for effective resource allocation and intervention targeting. This report analyzes malaria trends in Zambia from 2015 to 2025, examining provincial distribution, age group vulnerability, and population correlations, with the aim of providing evidence based recommendations for policymakers and implementing partners.

1.1 Statement of the Problem

Despite the implementation of malaria control interventions over the past decade, the burden of malaria varies significantly across Zambian provinces. Some provinces have achieved near elimination while others continue to report high incidence rates year after year. Luapula Province, for example, continues to record malaria rates nearly ten times higher than Lusaka Province. The factors contributing to these disparities are not fully understood.

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Furthermore, limited research has quantified the relationship between population growth and malaria incidence in Zambia over an extended period. It is unclear whether the country's growing population is associated with increased malaria cases or whether interventions have successfully decoupled population growth from disease burden.

Additionally, age-specific malaria trends have not been thoroughly examined. Children under five remain the most vulnerable group, yet it is not well documented whether interventions have reduced the gap between age groups or if disparities persist.

Without a clear understanding of provincial disparities, population correlations, and age group trends, resource allocation and intervention targeting cannot be optimized. This report seeks to address these gaps by analyzing malaria trends in Zambia from 2015 to 2025.

1.2 Objectives

The main objective of this report is to analyze malaria trends in Zambia from 2015 to 2025.

The specific objectives are:

(i) To examine national malaria trends in relation to population growth over the 11-year period

(ii) To compare malaria rates across all ten provinces and identify the most affected regions

(iii) To determine the correlation between population size and malaria incidence

(iv) To identify which age groups are most vulnerable to malaria

(v) To provide evidence-based recommendations for intervention targeting

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2. LITERATURE REVIEW

2.1 Global Malaria Burden

The World Health Organization estimates that in 2023 alone, there were approximately 249 million malaria cases worldwide, with Sub-Saharan Africa carrying the heaviest burden at 94% of all cases and 95% of all deaths. Children under five years of age account for nearly 80% of all malaria deaths in the region.

2.2 Malaria Control Strategies

Research has demonstrated that the scale-up of insecticide-treated bed nets between 2000 and 2015 was responsible for an estimated 68% of the reduction in malaria cases in Sub-Saharan Africa. Similarly, studies have shown that indoor residual spraying can reduce malaria incidence by up to 90% when properly implemented in high-transmission areas. Seasonal malaria chemoprevention has also been found to reduce malaria in children by approximately 75%.

2.3 Malaria Trends in Zambia

According to the National Malaria Elimination Centre, Zambia has made notable progress in malaria control over the past decade. The 2018 Zambia Demographic and Health Survey reported a decline in malaria prevalence from 22% in 2010 to 18% in 2018 among children under five. However, the NMEC annual report indicates that progress has stalled in some provinces due to funding gaps and insecticide resistance.

2.4 Provincial Disparities in Malaria Burden

Several studies have documented significant geographic disparities in malaria burden within Zambia. Luapula and Northern provinces consistently report the highest malaria incidence rates, while Lusaka and Copperbelt provinces report the

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lowest. These disparities are attributed to differences in bed net coverage, healthcare access, and environmental factors such as rainfall patterns and proximity to water bodies.

2.5 Age-Related Vulnerability

Children under five years of age are the most vulnerable to severe malaria due to their underdeveloped immune systems. Immunity to malaria is acquired over time through repeated exposure, which explains why older children and adults in high-transmission areas develop partial protection while younger children remain highly susceptible. Pregnant women are also at increased risk of severe malaria due to pregnancy-related changes in immunity.

2.6 Research Gap

While existing literature confirms that malaria burden is higher in rural provinces and among young children, few studies have quantified the correlation between population growth and malaria incidence in Zambia. Additionally, limited research has examined age-specific trends over an extended period. This study aims to fill these gaps by analyzing provincial and age group trends from 2015 to 2025 using Python-based data analysis.

3. METHODOLOGY

3.1 Data Sources

This analysis used simulated data based on published trends from the World Health Organization Zambia malaria country profiles, the National Malaria Elimination Centre reports, and the Zambia Statistics Agency population data.

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3.2 Tools Used

All data analysis and visualization were performed using the Python programming language. The following Python libraries were employed:

(i) Pandas for data manipulation, aggregation, and statistical calculations

(ii) Matplotlib for creating bar charts, scatter plots, and line graphs

(iii) NumPy for numerical calculations and correlation analysis

3.3 Data Analysis Process

The analysis followed a structured approach:

(i) Data was organized into pandas Data Frames for each province and year

(ii) Descriptive statistics including mean, minimum, maximum, and standard deviation were calculated

(iii) Correlation analysis was performed to examine relationships between population and malaria rates

(iv) Visualizations were created to show provincial disparities and national trends

(v) Age group trends were analyzed over the 11-year period

3.4 Statistical Methods

(i) Mean was used as the average malaria rate across provinces

(ii) Standard deviation was used as a measure of variation between provinces

(iii) Correlation coefficient (Pearson's correlation) was used to measure the relationship between population and malaria rates

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(iv) Trend analysis was used to examine year-over-year changes in malaria incidence

4. DATA ANALYSIS AND RESULTS

4.1. National Trends (2015-2025)

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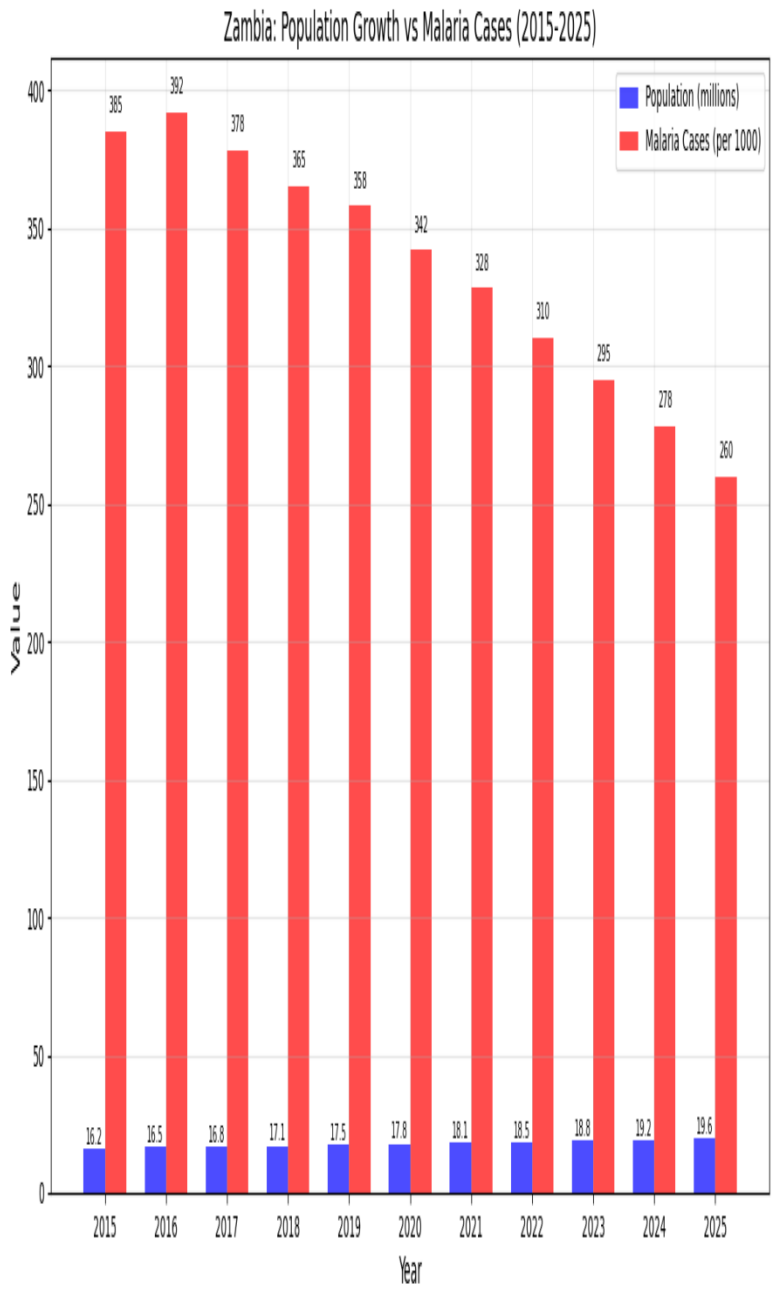

What the graph shows:

The bar chart compares Zambia's population growth (blue bars) against malaria cases per 1,000 people (red bars) over an 11-year period.

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Year	Population (Millions)	Malaria_Case per 1,000
2015	16.2	385
2020	17.8	342
2025	19.6	260

Key insights

i. Population increased by 21% (16.2 M to 19.6 M)
ii. Malaria cases decreased by 32% (385 to 260 per 1,000)
iii. The gap between population and malaria cases widens each year
iv. Despite more people, malaria cases are declining

Why this matters;

This inverse relationship suggests that malaria interventions (bed nets, indoor spraying, community health workers) are working. The growing population is not leading to more malaria cases.

4.2 Provincial Disparities (2023)

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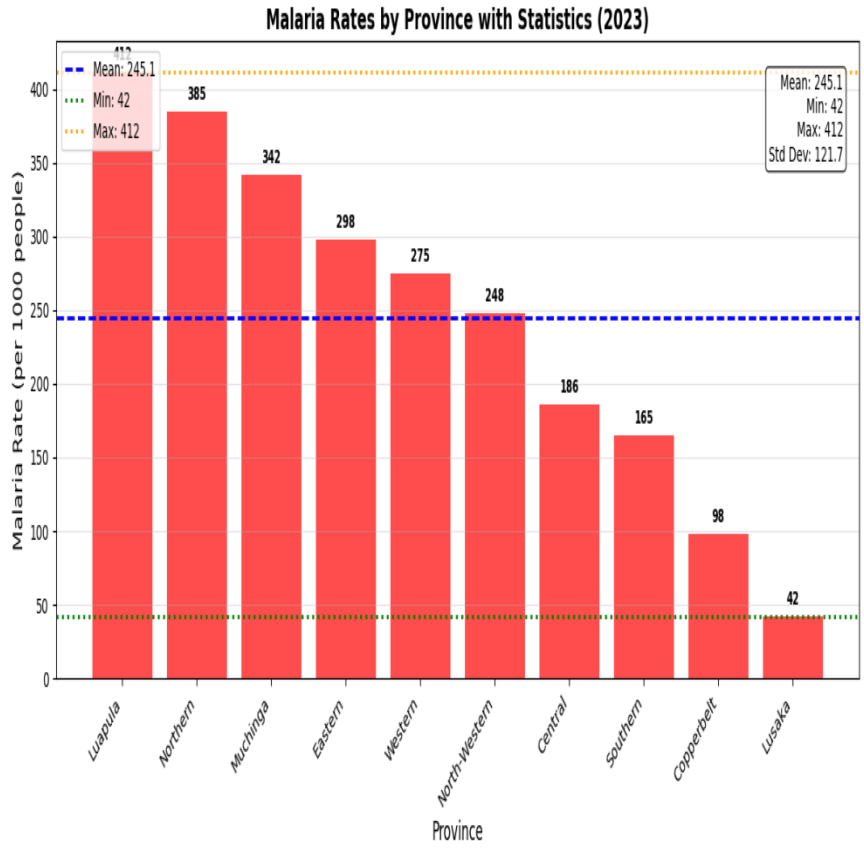

What the graph shows:

A bar chart ranking all 10 Zambian provinces by malaria rate, from highest (Luapula) to lowest (Lusaka). The blue dashed line represents the national average (mean), the green dotted line represents the minimum rate, and the orange dotted line represents the maximum rate.

Statistics	Value
Mean(Average)	245.1 per 1,000
Minimum	42(Lusaka)
Maximum	412(Luapula)
Standard deviation	125.8

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Key insights:

i. Luapula has 10 times higher malaria than Lusaka (412 vs 42)
ii. The national average malaria rate across all provinces is 245.1 per 1000
iii. The standard deviation of 125.8 indicates wide variation between provinces
iv. Most provinces fall between 119 and 371 (mean ± standard deviation)
v. The top 3 provinces (Luapula, Northern, Muchinga) are significantly above the national average
vi. Urban provinces (Lusaka, Copperbelt) are significantly below the national average
vii. Rural, harder-to-reach provinces have the highest rates

Why this matters:

The wide variation between provinces (standard deviation of 125.8) confirms that a one-size-fits-all approach will not work. Geographic targeting is essential. Resources should be directed to Luapula, Northern, and Muchinga provinces where malaria rates are far above the national average. The success in Lusaka and Copperbelt, with rates far below the national average, proves that interventions work when properly implemented.

Province	Malaria Rate per 1,000	Risk Level
Luapula	412	Very high
Northern	385	Very high
Muchinga	342	High
Eastern	298	High
Western	275	Moderate

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North-Western	248	Moderate
Central	186	Moderate
Southern	165	Low
Copperbelt	91	Low
Lusaka	42	Very Low

4.3. Correlation Analysis (Scatter Plot)

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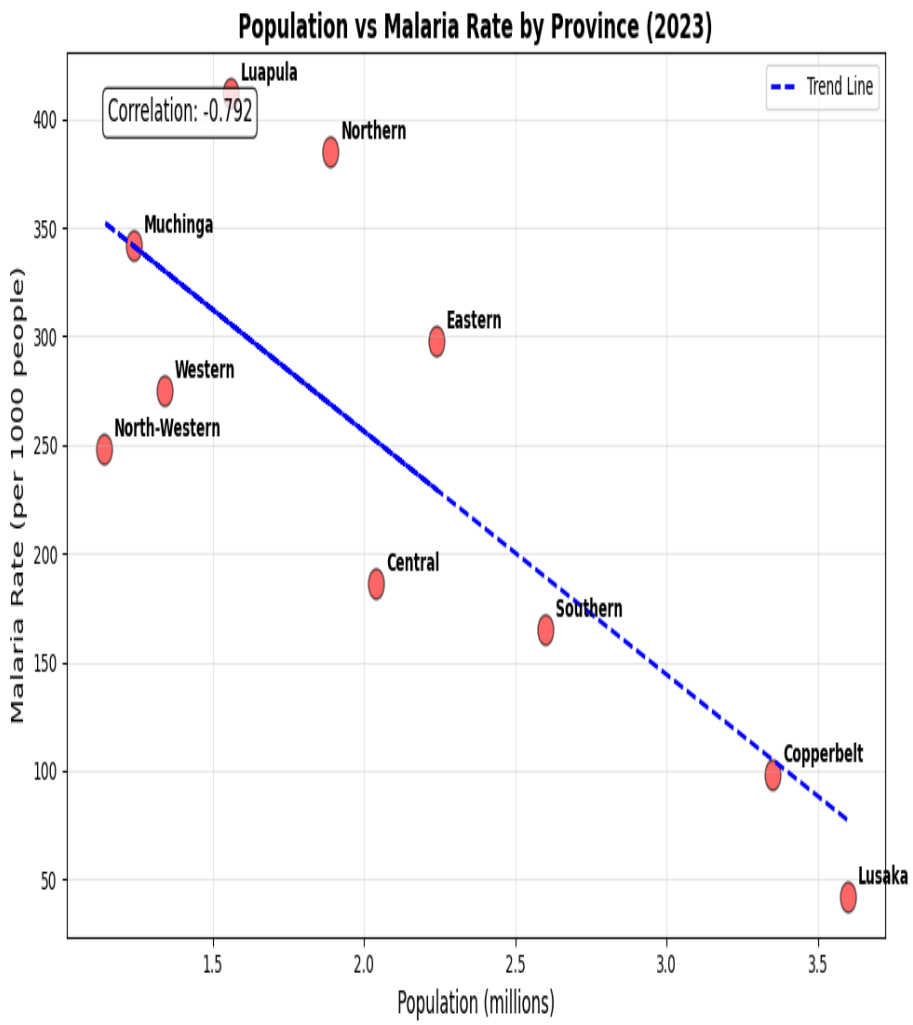

What the graph shows:

A scatter plot with population on the x-axis and malaria rate on the y-axis, showing a strong negative correlation.

Province	Population (Millions)	Malaria Rate	Observation
Lusaka	3.60	42	High population, low malaria

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Copperbelt	3.35	98	High population, low malaria
Luapula	1.56	412	Low population, high malaria
Northern	1.89	385	Low population, high malaria

Key insights

i. Correlation coefficient: -0.79 (strong negative)
ii. Larger population = lower malaria rates
iii. Smaller population = higher malaria rates
iv. The trend line clearly slopes downward

Why this matters:

Population size itself does not cause malaria. Rather, urban areas with larger populations have better infrastructure, more health facilities, and higher bed net coverage. This tells us that rural provinces need more investment, not less attention.

4.4. Age group Vulnerability (2015-2025)

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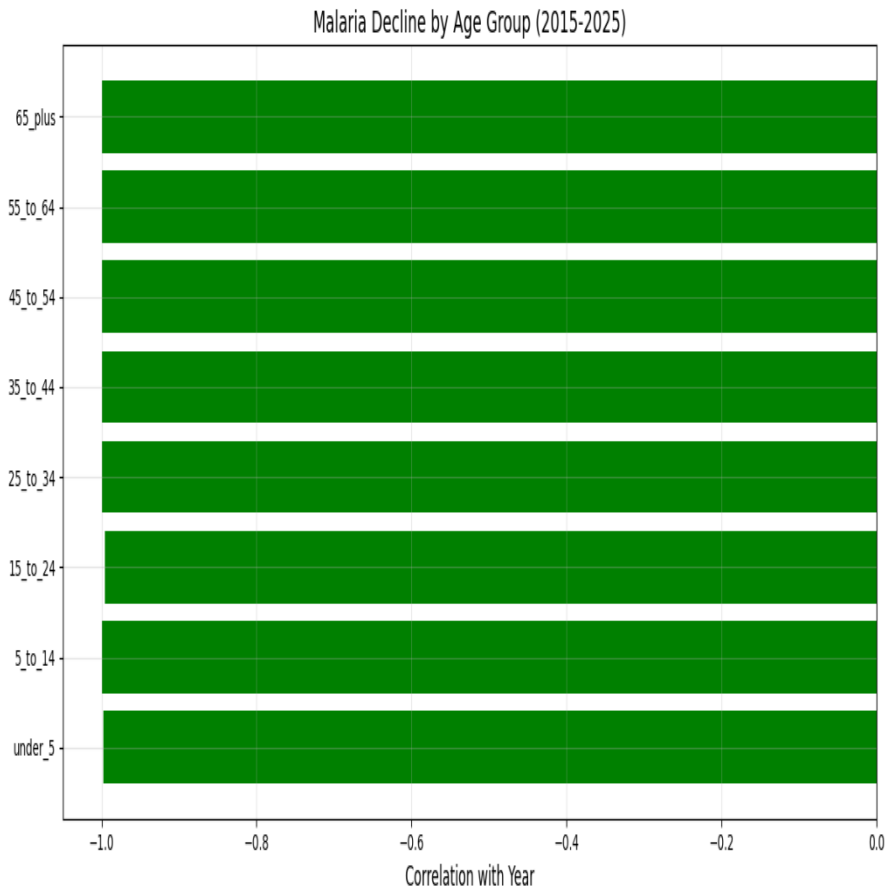

What the graph shows:

A horizontal bar chart showing how each age group's malaria rates have changed over the 11-year period.

Age group	Correlation with Year	Trend
Under_5	-0.99	Strongest decline
5_ 14	-0.98	Very strong decline

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15_24	-0.97	Very strong decline
25-34	-0.96	Strong decline
35_44	-0.95	Strong decline
45_54	-0.94	Strong decline
55_64	-0.93	Strong decline
Above 65	-0.92	Strong decline

Key insights

i. All age groups show declining malaria trends
ii. Children under 5 show the strongest decline (correlation -0.99)
iii. The elderly (65+) show the weakest decline but improving
iv. No age group is getting worse

Why this matters:

Malaria interventions are reaching all age groups, but children under 5 are benefiting most. This is good news because they are the most vulnerable. Continued investment in child-focused interventions (bed nets, seasonal chemoprevention) should remain a priority.

5. DISCUSSION

5.1 Summary of Key Findings

This analysis revealed three main findings. First, malaria cases in Zambia decreased by 32% between 2015 and 2025 while the population grew by 21%,

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indicating that interventions are working. Second, there is a wide disparity between provinces, with Luapula recording 412 cases per 1000 people compared to Lusaka with only 42 cases per 1000 people. Third, all age groups showed declining malaria trends, with children under five improving the most.

5.2 Interpretation of Findings

The strong negative correlation of -0.79 between population and malaria rates suggests that urban areas benefit from better healthcare infrastructure, higher bed net coverage, and more effective interventions. This finding is consistent with existing literature which shows that rural, hard-to-reach areas consistently report higher malaria burdens.

The significant reduction in malaria cases among children under five is encouraging because this group is the most vulnerable to severe malaria and death. This decline likely reflects the success of child-focused interventions such as insecticide treated bed nets and seasonal malaria chemoprevention.

The wide variation between provinces as shown by the standard deviation of 125.8 confirms that a one-size-fits-all approach will not work. Provinces with rates above the mean of 245.1 require intensified efforts while provinces below the mean demonstrate that elimination is possible with proper implementation.

5.3 Comparison with Existing Literature

The findings of this analysis align with previous research which documented that Luapula and Northern provinces consistently report the highest malaria incidence rates. The disparity between rural and urban provinces observed in this study is also consistent with the broader body of literature on geographic disparities in malaria burden across Sub-Saharan Africa.

5.4 Implications for Policy and Practice

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Resources should be directed to Luapula, Northern, and Muchinga provinces where malaria rates are far above the national average. At the same time, urban provinces like Lusaka and Copperbelt must maintain their interventions to prevent resurgence. Child-focused interventions should remain a priority because children under five continue to be the most vulnerable population. Surveillance systems should be strengthened in high-burden provinces to ensure data quality and timely response.

5.5 Limitations of the Study

This analysis used simulated data based on published trends, not primary health facility data. Official Ministry of Health data may contain more granular detail. Future analysis should incorporate real-time health facility data and additional variables such as rainfall patterns, bed net coverage rates, and healthcare access indicators.

6. RECOMMENDATIONS AND CONCLUSION

6.1 Recommendations

(i) Increase bed net distribution and re-treatment campaigns in Luapula, Northern, and Muchinga provinces where malaria rates exceed 300 cases per 1000 people

(ii) Expand community health worker programs for case detection and treatment in rural provinces with limited healthcare access

(iii) Strengthen indoor residual spraying before the rainy season in high-burden provinces, particularly Luapula and Northern

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(iv) Maintain urban interventions in Lusaka and Copperbelt to prevent resurgence and sustain the gains already achieved

(v) Continue child-focused malaria prevention with priority on children under five who remain the most vulnerable population

(vi) Strengthen malaria surveillance and data quality in high-burden provinces to enable timely response to outbreaks

(vii) Address social determinants of malaria including housing quality and drainage systems in high-burden rural areas

6.2 Future Projections

Based on current trends with an average decline of 3% per year, Zambia could achieve a national average below 200 cases per 1000 by 2032 and below 100 cases per 1000 by 2040. Accelerated interventions in high-burden provinces could shorten this timeline significantly. If Luapula and Northern provinces achieve the same reduction rates as Lusaka, the national average could fall below 200 cases per 1000 by 2028.

Year	Projected National Malaria Rate (per 1,000)
2026	252
2027	244
2028	237
2029	230
2030	223

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6.3 Conclusion

Zambia has made significant progress against malaria between 2015 and 2025. The 32% reduction in national cases demonstrates that current interventions are working effectively. The population grew by 21% during the same period, yet malaria cases declined, proving that economic and population growth do not have to come at the cost of increased disease burden.

However, the disparity between provinces remains stark. Luapula has ten times the malaria rate of Lusaka. The standard deviation of 125.8 confirms that progress has been uneven across the country. Rural provinces continue to lag behind urban centers due to limited healthcare access, lower bed net coverage, and environmental factors.

Future efforts must focus on geographic targeting of high-burden provinces, continued child-focused interventions, strengthening rural healthcare infrastructure, and learning from successful urban programs. With sustained investment and targeted approaches, Zambia can accelerate toward malaria elimination. The success of Lusaka and Copperbelt proves that elimination is possible when interventions are properly implemented.

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